parasite theory and Amotz Zahavi's 'handicap' theory. If you follow through the logic of my penis hypothesis, males are handicapped by the loss of the bone and the handicap is not just incidental. The hydraulic advertisement gains its effectiveness precisely *because* erection sometimes fails. Darwinian readers will certainly have picked up this 'handicap' implication and it may have aroused in them grave suspicions. I ask them to suspend judgement until they have read the next note, on a new way of looking at the handicap principle itself.

p. 207 … [Zahavi's] … maddeningly contrary 'handicap principle'

In the first edition I wrote: 'I do not believe this theory, although I am not quite so confident in my scepticism as I was when I first heard it.' I'm glad I added that 'although', because Zahavi's theory is now looking a lot more plausible than when I wrote the passage. Several respected theoreticians have recently started taking it seriously. Most worrying for me, these include my colleague Alan Grafen who, as has been said in print before, 'has the most annoying habit of always being right'. He has translated Zahavi's verbal ideas into a mathematical model and claims that it works. And that it is not a fancy, esoteric travesty of Zahavi such as others have played with, but a direct mathematical translation of Zahavi's idea itself. I shall discuss Grafen's original ESS version of his model, although he himself is now working on a full genetic version which will in some ways supersede the ESS model. This doesn't mean that the ESS model is actually wrong. It is a good approximation. Indeed, all ESS models, including the ones in this book, are approximations in the same sense.

The handicap principle is potentially relevant to all situations in which individuals try to judge the quality of other individuals, but we shall speak of males advertising to females. This is for the sake of clarity; it is one of those cases where the sexism of pronouns is actually useful. Grafen notes that there are at least four approaches to the handicap principle. These can be called the Qualifying Handicap (any male who has survived in spite of his handicap must be pretty good in other respects, so females choose him); the Revealing Handicap (males perform some onerous task in order to expose their otherwise concealed abilities); the Conditional Handicap (only high-quality males develop a handicap at all); and finally Grafen's preferred interpretation, which he calls the Strategic Choice Handicap (males have private information about their own quality, denied to females, and use this information to 'decide' whether to grow a handicap and how large it should be). Grafen's Strategic Choice Handicap interpretation lends itself to ESS analysis. There is no prior assumption that the advertisements that males adopt will be costly or handicapping. On the contrary, they are free to evolve any kind of advertisement, honest or dishonest, costly or cheap. But Grafen shows that, given this freedom to start with, a handicap system would be likely to emerge as evolutionarily stable.

Grafen's starting assumptions are the following four:

1. Males vary in real quality. Quality is not some vaguely snobbish idea like unthinking pride in one's old college or fraternity (I once received a letter from a reader which concluded: 'I hope you won't think this an arrogant letter, but after all I am a Balliol man'). Quality, for Grafen, means that there are such things as good males and bad males in the sense that females would benefit genetically if they mated with good males and avoided bad ones. It means something like muscular strength, running speed, ability to find prey, ability to build good nests. We aren't talking about a male's final reproductive success, since this will be influenced by whether females choose him. To talk about that at this point would be to beg the whole question; it is something that may or may not emerge from the model.

2. Females cannot perceive male quality directly but have to rely upon male advertising. At this stage we make no assumption about whether the advertisements are honest. Honesty is something else that may or may not emerge from the model; again that is what the model is for. A male could grow padded shoulders, for instance, to fake an illusion of size and